Schubert 多项式学习笔记

Contents

1 Z G _ " ” ¥ ü ’ ç l , ® © M : ` Q K Q p @ q , B M i 2 ç M

$$c^{w;d}_{u;v} = \frac{T}{?} (ev^T_1)^?((u)^T) (ev^T_2)^?((v)^T) (ev^T_3)^?(e(w)^T)$$

Ï $\overline{M}_{0;3}(X;d)$ ^ × ç ˜ ¥ b W ev_i ^ ß ˜ b
£ ü ± ^ S ` Q Q 7 a i ` i 2 , ; v

U R I V n b W X X # ö ° $B^0 = T$ (U U) ¥ T ¨ b

U k a ¥ ¨ g • T T ` Q D 2 + i B Q M | 7 Q Ž K m ̲ " H ¥ 9 Ø B ÿ ž X X ¥ 9 Ø f ™ b

U j V ¨ : ` ? K ¥ ž Ÿ ç Ø S ` Q T Q b B i B Q M e X R - (Ï : ` - V h ? K X X j X k)

è RXR U"`m?"ìm? i Q`¥2è`0 VX $S_3$ Ï 123 ^Klí 321 ^Kvíb $a_3$ ¥ "`m? i Q`/Ê½ T `iB H Q`/2> bbm2¹

$$\begin{tikzcd}[column sep=small]
 & & 321 \arrow[dll, dash] \arrow[drr, dash] & & \\
231 \arrow[dr, dash] & & & & 312 \arrow[dl, dash] \\
 & 132 \arrow[dr, dash] & & 213 \arrow[dl, dash] & \\
 & & 123 & &
\end{tikzcd}$$

'

$$123 < 213; \quad 123 < 132; \quad 213 < 321; \quad 132 < 321;$$

O 213 Ð 132 [# 231 Ð 312 "`m?½/ ,V1 BM+QKT `b# H2

çl RX92Ub+2M/i bÄ VXÂÐ w 2 S_n ¥ /2b+2M'ib çl¹

$$.2(w) = f\, i\, 2\, f\, 1; :::; n \quad 1g : w(i) > w(i+1)\, g$$

'¡@ $w(i) > w(i+1)$ ¥ÊÂ i ¥"† XÄñ /2b+2M|ÂÐs¹ÏÄv X

èÂ $w = 35142$ ' $(3;5;1;4;2)$ µ $w(1) = 3 > w(2) = 5$,îë $w(2) = 5 > w(3) = 1$ îë $w(3) = 1 > w(4) = 4$,îë $w(4) = 4 > w(5) = 2$ îë # $.2(w) = f\,2;4g$ X

çl RXa2UT ` i2/ .2b+2XM iu;v 2 $S_1$ bÏ K t .2(b) KBM (u;v) b5 (u;v) ¡@ b2T ` i2/ /2b+2bM ib

' u ¥Kv/†Ä,ÑV v ¥Kl/†Ä £ €¥/†u×,×ßb

RXj"Xµ Ø0 .BpB/2/ /Bz2`2M+2 QXT2` iQ`

çl RX"eµØ0 .BpB/2/ /Bz2`2M+2 QXT2¿ i=1;:::;n 1 "µØ0 .BpB/2/ @ çl¹

UkV
$$@(f) = \frac{f(x_1;:::;x_i;x_{i+1};:::;x_n) \quad f(x_1;:::;x_{i+1};x_i;:::;x_n)}{x_i \quad x_{i+1}}$$

÷Bî¹ ¿ÂÐ w çl ‹¥"µØ0 $@_w$ ¹a'F†¥ @X

è RXk'µØ0¥è0 VX S_3 Ï M ¹ $x_1;x_2;x_3$

$@(x_1) = \frac{x_1 \quad x_2}{x_1 \quad x_2} = 1$ b

$@(x_2) = \frac{x_2 \quad x_1}{x_1 \quad x_2} = \quad 1$ b

$@(x_1^2) = \frac{x_1^2 \quad x_2^2}{x_1 \quad x_2} = x_1 + x_2$ b

$@(x_1x_2) = \frac{x_1x_2 \quad x_2x_1}{x_1 \quad x_2} = 0$ b

$@(x_2^2) = \frac{x_2^2 \quad x_1^2}{x_1 \quad x_2} = \quad (x_1 + x_2)$ b

$@@(x_1) = @(1) = 0$ y¹ @ T¨¿ x_1 ¤ R @ T¨¿È"¤ yb

$@(x_2) = 1$ @ ŒÐx_2 „ x_3 b

"µØ0¥T¨^| ëÄ @ | [TM¹1¿ $x_i;x_{i+1}$ ¥ ë [Tb ç l R X d U 2 r" µ Ø 0 a F 2 r /B p B / 2 / /B z 2 ` 2 M + 2 Q W T 2 ¿ ` ñ Ø Â Ð

w; v 2 S_n a F 2 r" µ Ø 0 a F 2 r /B p B / 2 / /B z 2 ` 2 M + 2 Q T 2 ` $@_{w/v}$[k] ç l Q `

$$@_{w/v} = \sum_{\substack{u \\ v \leq u \leq w}} (-1)^{\ell(w)-\ell(u)} @_u \quad \text{U j V}$$

Ï p „ | R î µ ¡ @ v u w " ` m ? ½ ¥ Â Ð u b

ÿ i ƒ Ú , i $@_v^{-1}$ i i " µ Ø 0 @ ^ â , ¥ $@_v^2 = 0$ À µ I Ø 0 b a F 2 r" µ Ø 0 Y V F † ç l 7 d ¯ † I Ø 0 Ÿ ç l b

a F 2 r" µ Ø 0 ¡ @ [/ Ÿ É

- ' $v = e$ † Ê Â Ð H $@_{w/e} = @_w$
- ' $w = v$ H $@_{w/w} =$ B / š © Ø 0

è R X j U a F 2 r µ Ø 0 ¥ è 0 V X I n S_3 Ï ¥ + ñ 9 Ø P ¨ F † ç l $@_{w/v} = \sum_{v \leq u \leq w} (-1)^{\ell(w)-\ell(u)} @_u$

$@_{21/1}$ ƒ Ú $w = 21$ $v = 123$ † Ê Â Ð b ¡ @ 123 u 21 ¥ u º µ 123 „ 21 # $@_{21/123} = (-1)^{\ell(21)-\ell(123)} @_{123} + (-1)^{\ell(21)-\ell(21)} @_{21} = -@_1 + @_{21}$ b

$@_{231/12}$ $w = 231$ $v = 12$ š © Â Ð b ¡ @ 12 u 231 ¥ u µ 123; 213; 231 ¿ " ` m ? ½ # $@_{231/12} = @_{123} - @_{213} + @_{231}$ b

ç l R X 8 U ? m # 2 [T a + ? m # 2 ` i T Q H v M Q a + ? m # 2 [ T a + ? m # 2 ` i T Q H $\{S_w(x_1; \ldots; x_n) \mid w \in S_n\}$ ^ @ ™ a + ? m # 2 Ë ¥ i [T } V í ® G b + Q m a t + ? C i x 2 M @ # 2 ` ; 2 „ Æ b

ñ Ì V [Y V " µ Ø 0 @ n ç l R X ¾ w ç l

S H q ¿ K É Â Ð $w_0 = (n; n-1; \ldots; 1)$

$$S_{w_0}(t) = \prod_{i=1}^{n-1} x_i^{n-i} = x_1^{n-1} x_2^{n-2} \cdots x_{n-2}^2 x_{n-1}^1 \quad \text{U 9 V}$$

¾ w T ! i ^ w ¥ K ª / 2 b + 2 ' M @ $w(i) > w(i+1)$ ¥ K v i 5

$$S_w(t) = @_i(S_{ws_i}(t)) \quad \text{U 8 V}$$

Ï ws_i ^ | w ¥ » i Ê Ð » i + 1 Ê Œ Ð ¤ ž ¥ Â Ð b

n [T a + ? m # 2 [T r ^ Q ¥ n [K Ú Q [¹

$$S_w(t) = x_1^{c_1} x_2^{c_2} \cdots + \div ® Q ¥ [\quad \text{U e V}$$

Ï $c_i = \#\{j > i : w(j) < w(i)\}$ ^ w ¥ I ½ " _ b

è R X 9 U a + ? m # 2 ` T S_2 ¥ › œ 9 Ø V X S_2 Ï µ ñ Â Ð š © Â Ð 12[j] „ K É Â Ð $w_0 = 21$ b

K É Â Ð $w_0 = 21$ ® S H q $S_{21}(t) = x_1^{\binom{2}{2}} = x_1$ b

š © Â Ð $w = 12$ K ª / 2 b + 2 M i $w(1) < w(2)$ Œ $12 = 21 \, s_1$ #

$$S_{12}(t) = @_1(S_{21}(t)) = @_1(x_1) = \frac{x_1 - x_1}{x_1 - x_1} = 1$$

---

[k] Ù a F 2 r" µ Ø 0 Ð ô Y " µ Ø 0 µ I ¹ u Y $ n ¸ c b 2 + i B Q M

[j] ƒ Ú 12 ^ B › V U E Q M 2 @ H B M 2 V U Ç © Â Ð Q M 1 7! 1; 2 7! 2 ' $w(i) = i$, ^ b r T Â Ð (12) b

î [S_2 Ï

$$S_{12}(t) = 1; \qquad S_{21}(t) = x_1$$

è RX8 Ua+?m# 2`S$_3$ ¥›œ9Ø VX S_3 Ïµ $3! = 6$ ñÂÐbáÌP¨"µØ 0¾w„ ai MH2v[TZ7Ÿ9ØÄñ a+?m#[Tb

SHqÐ"µØ0 X ¿KÉÂÐ $w_0 = 321$ ® SHq

UdV $$S_{321}(t) = x_1^2 x_2.$$

"µØ0 @çl¹ $\partial(f) = \frac{f - s_i(f)}{x_i - x_{i+1}}$ Ï s_i ŒÐx_i „ x_{i+1} b

¿†[T @¥T¨?5¹

U3V $\partial(x_i) = 1; \qquad \partial(x_{i+1}) = -1;$

UNV $\partial(x_i^k) = x_i^{k-1} + x_i^{k-2}x_{i+1} + \cdots + x_{i+1}^{k-1}; \qquad \partial(x_j) = 0 \quad (j \neq i, i+1);$

URyV $\partial(1) = 0.$

¿É ¾w9Ø Xai2T $R(w) = 0$ š©ÂÐ

URRV $$w = 123. \quad S_{123}(t) = 1.$$

ai2T $\ell(w) = 1$

¿ $\ell(w) = 1$ ¥ÂÐ a+?m#[Ti® G2?K2ó`

URkV $w = 213 \quad (G2?K2(1,0,0)). \quad S_{213}(t) = x_1;$

URjV $w = 132 \quad (G2?K2(0,1,0)). \quad S_{132}(t) = x_2.$

ai2T $\ell(w) = 2$

¿ $\ell(w) = 2$ ¥ÂÐ P¨"µØ0 £b

$w = 231$:` bbK MMD(w) = f1g

UR9V $$S_{231}(t) = \partial(S_{321}(t)) = \partial(x_1^2 x_2);$$

UR8V $$\partial(x_1^2 x_2) = \frac{x_1^2 x_2 - x_2^2 x_1}{x_1 - x_2} = x_1 x_2;$$

UReV $$\Rightarrow S_{231}(t) = x_1 x_2.$$

$w = 312$ d :` bbK MMD(w) = f1g P¨ ai MH2v[TZ7 $S_{312}(t) = x_1 + x_2$ b

£ S_{312} ^ ë [Tb

ai2T $\ell(w) = 3$ KÉÂÐ

URdV $$w = 321. \quad S_{321}(t) = x_1^2 x_2 \quad \text{SHq} \quad .$$

S_3 a + ? m # 2 [ïT ø 9 V XÜ V • 9 Ø S_3 ¥ a + ? m # 2[T iÂ /

w	B › ” F	G 2 ? K 2 `	`(w)	$S_w(t)$

k X $w = 21$ K É Â Ð
N H $`(w) = 1$ ¡ @ u w ¥ µ $u = 12$ „ $u = 21$

$$S_{21}(t; v) = \sum_{u \quad 21} (\quad 1)^{`(21) \quad `(u)} S_u(t) \prod_{i=1}^{2} (x_i \quad y_{21(i)})$$
$$= (\quad 1)^{1 \quad 0} S_{12}(t) \quad (x_1 \quad y_2)(x_2 \quad y_1)$$
$$+ (\quad 1)^{1 \quad 1} S_{21}(t) \quad (x_1 \quad y_1)(x_2 \quad y_2)$$

Z 7 œ Ø

$$S_{21}(x; y) = \quad x_1x_2 + x_1y_1 + x_2y_2 \quad y_1y_2 + x_1^2x_2 \quad x_1^2y_1 \quad x_1x_2y_2 + x_1y_1y_2$$
$$= x_1^2x_2 \quad x_1^2y_1 \quad x_1x_2 + x_1y_1 + x_2y_2 \quad y_1y_2 \quad x_1x_2y_2 + x_1y_1y_2$$

Œ ÷ e ± ¥ 4 ³ ^ $S_{21}(t; v) = (x_1 \quad y_1)(x_2 \quad y_2)$ y ¹ $w_0 = 21$ ^ K É Â Ð ® ç l ° ¤ ó b

£ $S_{21}(t; v) = (x_1 \quad y_1)(x_2 \quad y_2) = x_1x_2 \quad x_1y_2 \quad x_2y_1 + y_1y_2$

ƒ Ð ë Z 7 ª † i] Ë [¥ ² T ^ B Á ¥ ë ¥ ¯ Z 7 ^ y ¹ á Ì Ü $\prod_{i=1}^{2} (x_1$ $y_2)(x_2 \quad y_1)$ 9 Z 7 L = $u = 12$ [Ï ‹ ¾ ¨ $w(1) = 2; w(2) = 1$ î [$\prod_{i=1}^{2}(x_i$ $y_{w(i)}) = (x_1 \quad y_2)(x_2 \quad y_1)$ b

è R X S_3 U ¥ 9 Ø V X S_3 Ï µ $3! = 6$ ñ Â Ð b á Ì G Q 9 Ø Ä ñ × a + ? m # 2 ` i T b

S_3 Ï ¥ Ü Å a + ? m # 2 T i ® $S_{w_0}(t) = x_1^2x_2$? ¨ " µ Ø 0 ¾ w

$S_{123}(t) = 1$
$S_{213}(t) = x_1$
$S_{132}(t) = x_1 + x_2$
$S_{231}(t) = x_1x_2$
$S_{312}(t) = x_1x_2$
$S_{321}(t) = x_1^2x_2$

C ¨ F † T $S_w(t; v) = \sum_{u \quad w} (\quad 1)^{`(w) \quad `(u)} S_u(t) \prod_{i=1}^{3} (x_i \quad y_{w(i)})$ 9 Ø b
R X $w = 123$ š © Â Ð
$`(w) = 0$ ¡ @ u w ¥ º µ $u = 123$

$$S_{123}(t; v) = 1$$

k X $w = 213$
$`(w) = 1$ $w$ ¥ " ` m ? ½ / Z í µ 123 „ 213

$$S_{213}(t; v) = (\quad 1)^{1 \quad 0} S_{123}(t) \prod_{i=1}^{3} (x_i \quad y_{213(i)})$$
$$+ (\quad 1)^{1 \quad 1} S_{213}(t) \prod_{i=1}^{3} (x_i \quad y_i)$$

® ¿ y M º C » B [„ » = [¥ ð Ï á Ì º ³ Z 7 - [

$$(x_1 \quad y_2)(x_2 \quad y_1) = x_1x_2 \quad x_1y_1 \quad x_2y_2 + y_1y_2$$

î [

$$S_{213}(t;\ v) = \quad x_1x_2(x_3 \quad y_3) + x_1y_1(x_3 \quad y_3)$$
$$+ x_2y_2(x_3 \quad y_3) \quad y_1y_2(x_3 \quad y_3)$$
$$+ x_1(x_1x_2x_3 \quad x_1x_2y_1 \quad x_1x_3y_2 + x_1y_1y_2$$
$$x_2x_3y_1 + x_2y_1y_2 + x_3y_1y_2 \quad y_1y_2y_3)$$

Œ ÷ e ± ¥ Z E ^ ° ¤ : þ $S_{213}(t;\ v) = x_1 \quad y_2$ ƒ ^ y ¹ 213 = s_1 ^ B ñ e † ¥ i ` M b T Q b B i B Q M

£ V F † i l $S_{213}(t;\ v)$ º µ B ñ / 2 b + 2 MÊiÂ R n [^ x_1 7 © ž [z h " î µ # y_2 µ | Ê Â ¥ [K Ô º : $x_1 \quad y_2$ b

L = Y V ° ¤ £ V ©

$$S_{213}(t;\ v) = x_1 \quad y_2$$

j Xw = 231

`(w) = 2 ¡ @ u 231 ¥ µ 123, 213, 231 ¿ É † ½ b á Ì G Q 9 Ø

$$S_{231}(t;\ v) = (\quad 1)^{2 \quad 0} S_{123}(t) \quad (x_1 \quad y_2)(x_2 \quad y_3)(x_3 \quad y_1)$$
$$+ (\quad 1)^{2 \quad 1} S_{213}(t) \quad (x_1 \quad y_1)(x_2 \quad y_2)(x_3 \quad y_3)$$
$$+ (\quad 1)^{2 \quad 2} S_{231}(t) \quad (x_1 \quad y_2)(x_2 \quad y_3)(x_3 \quad y_1)$$

9 Ø » B [$(x_1 \quad y_2)(x_2 \quad y_3)(x_3 \quad y_1)$

Z 7 $(x_1 \quad y_2)(x_2 \quad y_3) = x_1x_2 \quad x_1y_3 \quad x_2y_2 + y_2y_3$

ð [$(x_3 \quad y_1)$

$$(x_1x_2x_3 \quad x_1x_2y_1 \quad x_1x_3y_3 + x_1y_1y_3 \quad x_2^2y_2 + x_2y_1y_2 + x_2y_2y_3 \quad y_1y_2y_3)$$

» = [$x_1(x_1 \quad y_1)(x_2 \quad y_2)(x_3 \quad y_3)$

Z 7 $(x_1 \quad y_1)(x_2 \quad y_2) = x_1x_2 \quad x_1y_2 \quad x_2y_1 + y_1y_2$

ð [$x_1(x_3 \quad y_3)$

$$x_1(x_1x_2x_3 \quad x_1x_2y_1 \quad x_1x_3y_2 + x_1y_1y_2 \quad x_2x_3y_1 + x_2y_1y_2 + x_3y_1y_2 \quad y_1y_2y_3)$$

» Ø [$+ x_1x_2(x_1 \quad y_2)(x_2 \quad y_3)(x_3 \quad y_1)$

ÿ i ž » Ø [„ » B [™ T M] Œ " ” ,] b

Y V † i œ Ø K Ô V ¤

$$S_{231}(x;y) = x_1x_2 \quad x_1y_3 \quad x_2y_3 + y_2y_3$$

9 Xw = 312

Ë » 9 Ø V ¤

$$S_{312}(x;y) = x_1 \quad y_3$$

8 Xw = 132

Ë » 9 Ø V ¤

$$S_{132}(x;y) = x_2 \quad y_2$$

e Xw = 321 = w_0 K É Â Ð

® ç l ° ¤ ó

$$S_{321}(x;y) = (x_1 \quad y_1)(x_2 \quad y_2)(x_3 \quad y_3)$$

Z 7 ¹

$$x_1x_2x_3 \quad x_1x_2y_3 \quad x_1x_3y_2 \quad x_2x_3y_1 + x_1y_2y_3 + x_2y_1y_3 + x_3y_1y_2 \quad y_1y_2y_3$$

RX8]Ø Zm MimK +Q?QKXQHQ;v

çl RXR U]Ø Zm MimK +Q?QKQXHQX;v^;Á¯@™ $2\,H_2(X;Z)$
VUwLËb 0]Ø Zm MimK +Q?QKQHQH;v(X) ¥ðE ? çl¹
X

UkyV [ ]?[ ]= q h[ ];[ ];[ ]$i_{0;3;}$ [ ]

Ï

q ^™TM :cwLË¥ "
$hi_{0;3;}$ ^ :`QKQp@qBii2M :`QKQp@qBii2M BMp9"¡B@Mi
çHq¥•"ÄwLñ"

Ðô Y]Ø¥uY ôY]Ø $_u$ $_v$ = $\sum_w c^w_{u;v}$ $_w$ 7 0]Ø $_u$? $_v$ =
$\sum_{w;}$ q $c^w_{u;v}$( ) $_w$ uY ¿"" $c^w_{u;v}$( ) ÎG ¿õ¤0[¥wLË b
+… ÷ :`QKQp@qBMi2M þØÏ¥‰êÆ, ªŸ$ EQMib2@pB„+?
Æ}"+…bñ9"¥^{S:Ä¥×ç˜ ¥ñ" ^ôYMŒØ,¥ ówLñ' ób

RXe©M 0]Ø 1[mBp `B Mi +Q?QKáÌHçQ;v RXÏÜAž ×
a+?m#2Ti ‹¿ @™ ¥©M 0]ØË X '«áÌ|W%º ƒB,•À
QX

ÂTªôY¥]Ø^ùîbW¥ ó™ ó-*¹©M]Øü^ ùî ó{µ ëŸ
¥™ óXñ|+…Ù5 Ä¹}"Ù5 -^C} a+?m#2`i +¥H$mXHmb

çl RXRhQ`mÌbë VXhQ`mÌbë YÈ· C ¥µKð Å™T S^1 ¥
µKð ,

$$T = (C\ )^n \qquad T = (S^1)^n$$

°4 -áÌV[ÜñX`îB óõ• ó¥ð XèÂ-¹o¡¹à1 -ü^Bñ S^1
UR hQ`mTb¨Voë X

çl RXRskËbW ET *H bbB7vBM;VXT T=2(C)^n ¹Bñ n » iQ`mb
Ï C = Cnf0g VUd,¯"¥ðE• VU²5: * `i2bB M TbQNm+Ti
] ¿ n »¯Ìë +QKTH2t i9Q`[mTb T = $(S^1)^n$ ƒ^ n »LÌëb T ¥s
ËbW *H bbB7vBME;T^{3}T+ñ2‰êbW -¡@[/ ñ,•‹Ÿ ,
URWêŸ U*QMi` +iB,ETH2%‰ê ü^BñÄ -Àµ©… óó óX¨"ÐÔý
ª ,H (ET)=0 UîµÚ»ÿÄ]Ø¹, VX
Uk1V®T¨ U6`22 +iB,QMTVT¨ ET Àµ©…,îÄ X
¹l¹1„Æ ET\'áÌXpBñbW X • T T¨/¥ bW X/T ¥]ØH -
ÂTT¨,1® U tÄ È /,î V-bWöM¤dÈ óQ ó U¡ sÄ V•Á
.d¥]ØØ,>r X
"Q`2¥H?ŒXE^ , ;– X ¥T¨,1® -*üÜ X „Bñ ó' 1® ó¥bW
ET ð Ÿ 5y¹ ET ^Vê¥ -X ET ‰êŸÉ „ X ^B"¥ cŒy¹ T ET
T¨1® -î[T X ET ¥ ˜T¨9Bç^1®¥ X
°4 -ET V[Ø³¹îµÚ»oë¥ K ,

$$ET = \underset{m!1}{H}\,BSK^{2m+1}$$

---

[d]Ù ©M 0]ØË^l¹$n¸c b2+iBdQM

[3]Ù 1hsËbWµl¹°$Ø³$n¸c b2+iBNQM

Ï S^{2m+1} ^ $2m+1$ » o ë X
÷ 8 ¹ - ¿ $T = (C\ )^n$ - µ ,

$$ET = (S^1)^n = \underset{m!1}{H} B(S^{2m+1})^n$$

Ï S^1 ^ í k » o ë X

ç l R X R ç ! © M 0] Ø 1 [m B p ` B M i + Q ? Q K Q H T Q ÷ v $(C\ )^n$ ¹ B ñ i Q ` m b ^ B ñ { µ T T ¨ ¥ ‰ ê b W X b W X ¥ T @ © M 0] Ø ç l ¹ ,

UkRV $$H_T(X) = H\ (X\ \ _T ET)$$

Ï ET ^ T ¥ s Ë b W U ç l R X R k $_T$ ET ^ T T ¨ / ¥ b W - ë ¹ " Q ` 2 H / " Q ` 2 H + Q M b i ` m + i] B Q M ? Q K Q i Q T v [m Q X B 2 M i
© M] Ø $H_T(X)$, Ç Ç ^ B ñ Ì - Î ^ B ñ U K Q / m Î µ 2 ¥ © M] Ø Ë V [A T ^ [[T ¹ " " ¥ } " ` X

è R X ® © M 0] Ø ¥ è 0 V X Ä b W pt ,

$$H_T(pt) = H\ (BT) = Z[y_1; : : : ; y_n]$$

Ï y_i ‹ $T = (C\ )^n$ ¥ Ä ñ 3 î í ¥] Ø Ë X
+ Y ¹ - ¿ $T = S^1$ -BT = CP^1 U k » ¯ • b W V µ ,

$$H_{S^1}(pt) = Z[y]$$

Ï y ^ CP^1 ¥ » B * ? 2 ` Ë X
¿ $T^2 = S^1$ S^1 -BT ^{2} = CP^1 CP^1 - µ ,

$$H_{T^2}(pt) = Z[y_1; y_2]$$

@ ™ $Fl_n(C)$, $H_T(Fl_n(C))$ T ¹ $H_T(pt) = Z[y_1; : : : ; y_n]$ @^ 1 ® ¥ - ¸ ®
a + ? m # Ë 2 f [X (w)]$_T$ j w 2 S_n g ó X ƒ i Å " á Ì V [Y V } " m U Ø [
T V Ÿ Ú ' 9 Ø @ ™ ¥ + … Œ Ä Ù 5 X

ç l R X R a 9 + ? m # 2 Ë i a + ? m # 2 ` i + H X b @ ™ $Fl_n(C)$ V [ $ s ³ ¹ ¸ l ¥ + … † í - ë ¹ a + ? m # 2 ` i a + ? m # 2 ` i + X Ä ñ H 8 ¥ > ë ¹ a + ? m # 2 Ö Õ a + ? m # 2 ` i p ` B X 2 i v
a + ? m # Ë 2 û ^ ƒ t Ö Õ] Ø , Ï î ‹ ¥ $ Ë X ñ Ì ü ^ ^ @ ™] Ø ¥ óð 0 ó óS ¸ ó X © M] Ø Ï - Ä ñ a + ? m # Ë 2 û i ‹ B ñ × a + ? m # 2 T i X

© M 0] Ø | . d ¥] Ø Ð i Q ` m ¥ T ¨ M ² † - Á 3 ~ ¶ ¥ } " ² X ñ P ¤ á Ì V [Ü ¯ ¥ + … Œ Ä 9 Ø Ù 5 Ä ¹ [T Ì ¥ L Ÿ } " Ù 5 X

R X d X : ` b b K M Ð M : B b M b K M M B † M X

ç l R X R : 8 b b K M M B X : M b b K M M G r(k;n) ^ n » ¯ _ b W Ï î µ k » 0 b W ¥ " † X ñ ^ @ ™ ¥ + è ¡ @ V_1 V_2 V_k ¥ ~ ; b X

ç l R X R : è b b K M M B † M X ë Â Ð w 2 S_n ¹ k @ : ` b b K M † M B Â M w ¥ I ½ " ¡ @

UkkV $$w(1) < w(2) < \quad < w(k) \quad O \quad w(k+1) > w(k+2) > \quad > w(n)$$

' - k ñ Ê Â ¾ 9 ª n k ñ Ê Â ¾ h X
k @ : ` b b K M † M B M G r(k; n) Ï ¥ a + ? m # 2 ` i B ‹ X

è R X N U : ` b b K † M ¥ M è B M X S_4 Ï

$k=2$ $w=(1;3;4;2)$ - kÊ 1 < 3 ¾9 ª kÊ 4 > 2 ¾h #^ k@
:` bbK MMB M

$k=2$ $w=(2;4;3;1)$ - kÊ 2 < 4 ¾9 ª kÊ 3 > 1 ¾h #^ k@
:` bbK MMB M

$w=(1;2;3;4)$ š©ÂÐ ^ $k=4$ ¥ :` bbK MMîBµÊMÂû¾9b

$w=(4;3;2;1)$ KÉÂÐ ^ $k=0$ ¥ :` bbK MMîBµÊMÂû¾hb

$w=(1;4;3;2)$ - kÊ 1< 4 ¾9 Œª kÊ 3 ,v¿ 2 #,^ :` bbK MMB M

:` bbK MM†BI'M :` bbK MM@B™Mr(k;n) Ï a+?m#8¥iÊÂ• X

:` bbK MM†B^M:` ? KðS TQbBiÇØBîî¥ `b

RX3VÈ žÙ}" áÌ¹I¹³1ñ$ X '«Ï áÌº £üî³¥v†s}
"ý b ƒ†sH €V?öÙ ¹I¹1„Æƒt `ÀQ$ñÌV…7Ÿ$

Bñ º¥Ù5 Â…ùîõÃ ë$

Vx˜žÚ® "ÐEÌB° È „ ë"ÙbX`Bº bÏ1®È ¥Tii
ñ¥ÄBñÔ_û ‹bWÏ¥BñÈ bƒtÈ V[†î 5_P _ V[
IHŒ¹Üí IÈ bƒ î Bñ•ii +yžŒ• $SO(3)$ b

ŒƒÚµñ'Y¥+…Ù5 $SO(3)$ ^Bñ•w¥wë k@ë S^2 ¥MW ÄBÄ
)û,^Ü¥báÌÂ…¨LŸ}" Ü bW Ÿùî•w¥ `$

aQT?mb¥G?BŒXE R3dMy} ^ Ð ùîœñ• ,Âùîñ BñÄ¸Í
¥LŸÍ»b 8Ÿª |†Êí)¥MbWiiƒó •¥ †LŸÄb

V•ž}" "è0

žáÌVKe†¥ffƒ7Sb

è RXR=y»ÈE • $SO(2)$ VXÜë ¥È îBñ•bÄñÈ V[•îBñ 2 2
"

$$R( ) = \begin{pmatrix} +Qb & bBM \\ bBM & +Qb \end{pmatrix}$$

Ï ^È ˜ b

' $=0$ H¤ž†Ê " IbC Ù †Êí¸Í È "Â…MÄ$
p•i $=0$)|´

$$\frac{d}{d} R( ) \Big|_{=0} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} =: J$$

ƒñ " J } óíkIÈ ó¥• bÿi $J^2 = I$ ƒiÅ"Q¯@FíkIÈ
öíž Äiiü^ õ <B•b

îµ™Â J ¥ " îBñ_ bW » ¹ Rbƒü^ $SO(2)$ ¥Ù}" :¹ $so(2)$
$so(2) = R$ b

è RXRØR»ÈE • $SO(3)$ VXØ»bWÏ¥È ÷~¶bÄñÈ " $R \in SO(3)$ ¡
@$R^T R = I$ O /2 $R=1$ b

$SO(3)$ ¥Ù}" $so(3)$ ®îµ 3 3 Q ë " î

$$so(3) = \{X \in R^{3 \times 3} \mid X^T = X\}$$

¹I¹^Q ë$y¹†Ê ")¥MbWA¶<°¿ÿ•wë $R^T R = I$ ¨+…¥
ÅŸª MbWžŒ¿0 b

so(3) ¥ » ^ j i i ‹ Ø ñ U S à ¥ í k l È b S ¹

$$J_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}; \quad J_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}; \quad J_3 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

ñ Ì ¡ @ 1 " Ù “ |

$$[J_i; J_j] = J_i J_j \quad J_j J_i = \quad {}_{ijk} J_k$$

Ï $_{ijk}$ ^ G 2 p B @ * B p f B i Ô á Ì 5 ¡ i à ¡ j à È Ð 5 ¡ j à ¡ i à È ² T ,] i i µ s ® » Ø ñ à ¥ È Y Â b

V ” ž ` Ù } ” ¥ Ø Ä

C á Ì Ü 4 ³ ž ¥ T 4 ÿ Ÿ b

ç l R X R Ù } U V X B ñ Ù } ” g ^ × R C ¥ _ b W { µ B ñ = í Ø

$$[;] : g \quad g ! \quad g$$

ë ¹ Ù “ | ¡ @

U R V L Ÿ $[ax + by; z] = a[x; z] + b[y; z]$ $[z; ax + by] = a[z; x] + b[z; y]$

U k Q V ë Ÿ $[x; y] = \quad [y; x]$

U j V C + Q # $ ® T $[x; [y; z]] + [y; [z; x]] + [z; [x; y]] = 0$

¹ l ¹ Q ë $ y ¹ ” f ™ $[X; Y] = XY \quad YX$ ž z ^ ” ð E ¥ b

¹ l ¹ C + Q š # © B T $ ƒ Ÿ 1 ² † p $A(BC) = (AB)C$ ¥ ÿ • i i ñ £ V ” • ? õ Ÿ ¥ ² Ÿ É b

è R X R E k î l U Ÿ } ” gl_n V X î µ n n ” î B ñ Ù } ” Ï

$$[X; Y] = XY \quad YX$$

ƒ ë ¹ $gl_n(R)$ $gl_n(C)$ b ñ ^ K ó v ó € ¥ Ù } ” i i À µ © … ÿ • b

' Ó Ï $G = \ :G_n(C)$ ¥ Ù } ” ž ^ $g = gl_n(C)$ b

0 } ” Ð â í @ ™ ¥ + …

C á Ì í ž @ ™ $Fl_n(C)$ ¥ ù î b ¹ l ¹ @ ™ × 1 $ y ¹ ñ ^ ù î B î L Ÿ • $GL_n(C)$ ¥ 1 – ÷ û b

! $G = \ :G_n(C)$ b ñ ¥ 0 • ² ó á Ì @ ™ ¥ + …

ç l R X R 3 @ U ™ ¥ 0 • V X ! $G = \ :G_n(C)$ á Ì l n [/ × 1 0 •

B Ø ˜ V I ” " Q ` 2 H B • " Q ` 2 H b m # ; ` Q m T

B / Ø ˜ V I ” M Q " Q ` 2 H B • P T T Q b B i 2 " Q ` 2 H b m # ; ` Q m T

T ˜ ” •] ¿ $(C)^n$ v Ì ë J t B K H i Q ` m b

N † Ê Ø ˜ ” B ¥ â ô l M B T Q i 2 M i ` / B + H

N † Ê / Ø ˜ ” B ¥ â ô l M B T Q i 2 M i ` / B + H

ÿ i T = B \ B O N] ¿ @ ™ † Ê í) ¥ M b W b

¹ l ¹ @ ™ @ ™ Ø , Ï Â N × 1 $ y ¹ G/B ¥ + … Q ˜ G 1 & ¥ + … b @ ™ ^ G ¥ Ÿ b W i i F V [Ü G/B A T G Ï ó € Õ ó B ¥ b

â ” ¥ + … N Ð @ ™

N Ï¥Ä ñ í Í û V [• î · ” ˜

$$2\,t(X) = I + X + \frac{X^2}{2!} + \quad (X\ ^1 ø ì / Ø ˜ \ ”)$$

ƒ ó V $n = G(N2)$ ž N ¥ b
N ^ õ Y ¥ · ” ˜ ^ ; Á ¥ O » ” ¹ $\frac{n}{2}$ i i z © ¿ @ ™ ¥ » ” +
… N] ¿ @ ™ $Fl_n(C)$ ¥ 7 8 a + ? m # 2 3 ` b

è R X R $n = 2$ ¥ f ™ V X’ $n = 2$ $Fl_2(C) = CP^1$ ¯ • ° L b N H

$$N = \begin{matrix} 1 & 0 \\ a & 1 \end{matrix} : a\,2\,C = C$$

7 CP^1 † ž ^ C € Õ B Ä ª ¥ o ë] ~ ¿ k » Ü ë b

* ` i M 0 } ” Ð ô " d Â … s ³ _ b W
¹ Ø ³ Ù } ” ¥ ² á Ì ³ 1 B ñ z ¥ U S " b

ç l R X R * N ` U M 0 } ” \* ` i M b m # H ; 2 ¥ X! $g = gl_n(C)$ b î µ ˜ ”
î ¥ 0 } ”

$$h = f\ /\ B\ (t_1; : : : ; t_n) : t_i\ 2\ Cg = C^n$$

^ B ñ v ¥ - : 0 } ” i i © … ñ ˜ ” û ^ b ƒ ñ h ë ¹ * ` i M 0 } ”
* ` i M b m # H ; 2 b # `

¿ B î ÿ Ä • * ` i M 0 } ” ç l ¹ v ¥ - : 0 } ” b gl_n Ï h ž ^ î µ Ð
1 & Œ Ð ¥ ” î ¥ 0 } ” b

ô " d Ü ¯ Ù } ” s ³ ¹ e † Þ Ë
¿ gl_n á Ì 1 • ¥ ^ ô Û V U / h ¥ T ¨ b ç l ƒ r 2 B ; ? i

$$H\ E_{ij} = (t_i\ \ t_j)E_{ij} \quad Ï\ H = /\ B\ (t_1; : : : ; t_n)$$

ƒ ó + ˜ ´ s ³

$$gl_n = h \bigoplus_{i \neq j} C\ E_{ij}$$

÷ B î ¹

ç l R X k ô " U _ Q Q i b v b i 2 ¥ X! h ¹ * ` i M 0 } ” b ç l

$$= f\ \ 2\ h\ j\ 9X \neq 0; [H; X] = (H)X\ \ î µ\ \ H\ 2\ hg$$

¹ h ¥ ô " d _ Q Q i b v b i 2 Ä ñ 2 ë ¹ B ñ ô ` Q Q i ‹ ¥ X 3 î ¥ 0 b
Wg ^ B » ô b W b
gl_n ô ^ $_i$ $_j$ $i \neq j$ Ï $_i(H) = t_i$ ' ˜ ” ¥ » i ñ ˜ í b á Ì µ
s ³

$$g = h \bigoplus_{2} g$$

Q Ð q 2 v H • µ K ë ¥ Ÿ ÷
ô " d µ B ñ Å ó ¥ Ÿ É ñ Ì ® B F ó ' Q ó 3 î b

ç l R X k (R 2 U H • q 2 v H ; ` Q m) V X ¿ Ä ñ ô 2 ç l Ñ Ü ë H = f H 2 h : $\ell(H) = 0$ g b 1 ¿ ƒ ñ Ñ Ü ë ¥ Q s T ¨ h b

® î µ Q s 2 3 î ¥ µ K • ë ¹ q 2 v H • q 2 v H ; ` Q m: T W b

g l$_n$ W = S_n i i Â Ð • Ä ñ Â Ð û ^ t ' Q ¥ ð b ° 4 S_n í • U S à - W ¥ ë Ÿ b

g l$_n$ Ï e † Q s_i º Œ Ð M # ¥ i „ i + 1 U S à b ƒ Ð * Q t 2 i 2 ¥ ² › † B Á b

l ½ " q 2 v H • í ¥ Ê Â l '

ó ç B ñ q 2 v H • í w 2 W = S_n ñ Â … ó T ¨ ó ô " A $

ç l R X k (l ½ " A M p 2 ` b B Q M d b 1 2 l ") V X ¿ w 2 S_n ç l

l ½ " A M p 2 ` b B Q M $I(w) = f(i;j) : 1 \quad i < j \quad n; w(i) > w(j) g$

d l ½ " $J(w) = f(i;j) : 1 \quad i < j \quad n; w(i) < w(j) g = I(w_0 w)$

Ï $w_0 = (n; n \quad 1; :::; 1)$ ^ K É Â Ð b

+ … $I(w)$: c w Ü ' t ¨ ½ ó J ó b $jI(w)j = `(w)$ ž z ^ w ¥ É " ` m ? ½ Ï ¥ Ö b

0 • N (w) V @ ™ ž @ ™ ¥ ¡ ^

K ª á Ì º @ ™ + … Ï ¥ 1 o 0 • b

ç l R X k (0 • U N (w) Ð B (w)) V X Ä ñ w 2 W ç l

$$N(w) := N \setminus wNw^{-1}$$

"

$$B(w) := T \quad N(w)$$

+ … N (w) ^ N ¥ B ñ 0 [» " ¹ $jI(w)j$ b

ô > m K T ? ` 2 N v b a 2 [ + i B Q T M ¹ [ k 3 ) N (w) ] ¿ a + ? m # 2 B ` i B wB/B = $C^{`(w_0)} \quad ^{`(w)}$ Y V x 7! xwB/B b + Y ¹ N (w) ^ õ Y ¥ b

k X„ ý

' á Ì ù î @ ™ F l$_n$(C) ¥] Ø Ì H B ñ ' Ù 5 ^ ñ a + ? m # Ë 2 M i ð ö ¤ ž l ¹ $ ÷ Ú ' ¹ ª Â T á Ì ¨ × a + ? m # 2 [T i $S_w(t; v)$ T ¹ @ ™ © M] Ø Ë ¥ } V í [N] * ¹ ð $S_u(t; v)$ $S_v(t; i)$ Â … Z 7 ¹ † ñ $S_w(t; i)$ ¥ L Ÿ F † $ Z 7 " " $c^w_{u;v}(v, i)$ ¡ @ l ¹ Ÿ É $

ƒ „ • á Ì ù î [/ Z 7 T

$$S_u(t; v) \quad S_v(t; i) = \sum_{w \in S_1} c^w_{u;v}(v, i) \quad S_w(t; i)$$

Ï $c^w_{u;v}(v, i)$ ^ 1 ¿ v „ i ¥ [T b [R y]

[N] × a + ? m # 2 [T i ^ G b + Q m t @ a + ? C Ë x ¥ 2 1 M + w 2 < ; „ 2 Æ - " ¥ v M Ÿ : c @ ™ ¥ + y Ä • b

[R y] v= i = 0 H $c^w_{u;v}(0;0)$ 2 N ^ Ü Å ¥ a + ? m # ² 2 Ë " 9 " @ ™ Ï a + ? m # 2 B ` ¥ Œ Ä ñ " i i ' > B H # » 2 Ë i ô Ù 5 n b 2 + i B R Q k ¥ M, • b kXRVX+yžBî Øñ‰X¥ Ä Xùîƒ ñÙ5¥Bñ1–¦{^VKe†¥f ƒ7S –ªï„w<b

»B„ :` bbK MMB'™Mb ' u „ v û^ :` bbK MM†BHMYfM¤+Y b b k@:` bbK M†MBwM¥ a+?m#2[TiV[•îLŸy0¥ð $S_w(t; v) = Q_{i=1}^{k}(x_{a_i} \; y_{b_i})$ ƒÕy0Ä™TP¤ð Z7¥² B" –b JQH2p@ak) nMQ( ƒt""¥F† T EMmibQMR@0n Q+(‥³db ƒÕ ß/ :` ? K3) £ü [/žŸ²T

çØ RXR U:` ? K SQbBiBpB3-iv h??2Q`2KX(j!(X k; v)V2 S_n sY^ k_1@ :` bbK MM„B$k_2$M@:` bbK M†MB5 M

$$c^{w}_{u;v}(v; i) \, 2 \, N[t_i \quad y_j]_{i;j \quad 1}$$

'Z7""^1¿ t_i y_j ¥dµœ""" [Tb [RR]

»=„ €Õ :` bbK MMBKÅMb Bñ1–¥Ù5^ :` ? K¥žŸ²T?Žw <ž©i† $ a Km2kHR4([/‰Xii''Ó¥ö1²T h?2Q`2K RXR

çØ RXk Uh?2Q`2K R%RX,(kaR K(mM2 H2+imX!2u;R;XvR2)S1 5 $c^{w}_{u;v}(v; i)$ 2 $N[t_i \quad y_j]_{i;j \quad 1}$ b

´¤ÿi¥^ " •f™" b2T ` i2/ /2b'+2KM itb 2(b) KBM (v2)bnç l RXf9™9$ sùîb a Km2kHRy(ë žŸ T 6 M4:mQ„»=T€ (8) P¨ TmxxH8b :`Qi?2M/B[2T¥FS¶=ó 6Bñ Tb

»Ø„ Ke†¥+èb ÂTáÌ a Km2%HXÏ| v= i=0 5""|Ä¹ôY dµœ"bNHZ7TM¹

$$S_u(t; y) \quad S_v(t; y) = \sum_{w} c^{w}_{u;v}(0;0) S_w(t; y)$$

ƒž^ EB`B%HXQp, RX'R *Q`QHH ìv•R¥X™K

w, RXR U*Q`QHH `v R%X K,(REyB`QBMHDH2Q+piXn`¿2 R;y)Vw 2 S_n µ $@_{w/v}S_u(t)$ 2 $N[t]$ b

ƒÚ $@_{w/v}$ ^ J +/QM RHy„(Æ¥ bF2'rµØ0 Ð 6QKBM@E}B`B(e)HáHMQMb 1b

ÂmîU Øñ‰X¥HqGQhÐiiV1p u;v û^ :` bbK MMž©iM žKe†¥ v= i=0 f™b

‰X¥ªQ	Hq	}VçØ
»B„	u;v û^ :` bbK MMB M	:` ? K SQbBiBpBiv
»=„	u;v ©i†	a Km2%HX
»Ø„	v= i=0	EB`B%HXQp

'Ó¥£üG ¿ÚÄñ'¥ :` ? KžŸçØ çØ RXjyë "" $c^{w}_{u;v}(v; i)$ ¥ •+…³d [Rk]T¹¬ÁÖÎó "BHHT2v(j)ÏžŸ¥+…³db

áÌ[1¿ :`Qi?2M/B[2T¥F‰X²•'Ó¥), n b2+iB3XQ8M

kXkçXØ¥è0 X¹úù €yë°4†` áÌGQ £ØñçØ¥žŸ², Ä „û¸{W%9Øb

[RR] U ¹l¹ :` ? K SQbBi1BpBìvbbK MM†BMk_1; k_2$ ¥T¨^l¹$n¸c b2+iBk0M

[Rk] ƒÐ EMmibQMR@0n Q³(d,]iiWn¸c b2+iBR0M k X j X :` ? K S Q b B ì B q k B ‹ i v X

è R X R ç 9 Ø U R X R ñ :` b b K M M† B ¥ ð M V X¹ I¹ :` b b K M M B† M+ y $ Â - î • k @ :` b b K M† M B w M¥ a + ? m # 2 T i $S_w(t; v)$ V [• î L Ÿ y 0 ¥ ð

$$S_w(t; v) = \prod_{i=1}^{k} (x_{a_i} \quad y_{b_i})$$

ƒ Õ y 0 Ä ™ T P ¤ ð Z 7 ¥ ² + Y b i i á Ì º ³ | F L Ÿ y 0 M ð – ª × • œ Ø ' V b

8 è 0 $u = 21$ $v = 12$ 21 ^ :` b b K M M B 1 M ^ š © Â Ð b

° V ¤ a_3 Ï × a + ? m # 2 T i n è R X d

$$S_{21}(t; v) = x_1 \quad y_1; \quad S_{12}(t; i) = 1$$

9 Ø ð

$$\begin{aligned} S_{21}(t; v) \quad S_{12}(t; i) &= (x_1 \quad y_1) \quad 1 \\ &= x_1 \quad y_1 \end{aligned}$$

N H Z 7 " " $c^{21}_{21;12} = 1$ 2 N ƒ ^ B ñ Ü O ¥ ž Ÿ £ b

÷ B î ¥ è 0 $u = 132$ $v = 213$ ñ û ^ :` b b K M M B M

° V ¤

$$S_{132}(t; v) = (x_1 \quad y_2)(x_2 \quad y_2); \quad S_{213}(t; i) = x_1 \quad t_1$$

9 Ø ð

$$\begin{aligned} &S_{132}(t; v) \quad S_{213}(t; i) \\ &= (x_1 \quad y_2)(x_2 \quad y_2)(x_1 \quad t_1) \\ &= (x_1x_2 \quad x_1y_2 \quad x_2y_2 + y_2^2)(x_1 \quad t_1) \\ &= x_1^2x_2 \quad x_1x_2t_1 \quad x_1^2y_2 + x_1y_2t_1 \\ &\quad x_1x_2y_2 + x_2y_2t_1 + x_1y_2^2 \quad y_2^2t_1 \end{aligned}$$

U Z 7 V

æ ¨ $t_1 = (t_1 \quad y_1) + y_1$ | c t_1 ¥ [· s V £ î µ " " û ^ $t_j \quad y_i$ ¥ d µ œ " " " [T b [Rj]

² , ' $u; v$ û ^ :` b b K M M† B H M ð Z 7 " " ' L û ^ $(t_j \quad y_i)$ ¥ d µ œ " " " [T ƒ £ ç Ø R X ¥ R ý b

k X 9 X a K % m X H H q h Ð € Õ :` b b K M M B 1 p M X

è R X R ç 8 Ø U R X d k :` b b K M M† B ¥ ð M V X ç Ø R X 2 k ë ' P $u; v$ , ^ :` b b @ K M M B† M Z 7 " " ¯ – û ž Ÿ b ¹ ª ü ƒ B Ä á Ì I n B ñ d :` b b K M M B M ¥ è 0 b

d :` b b K M M B† M $u = 231$ ¥ ² $u = 231$ ¥ " F ¹ $(2; 3; 1)$ µ ñ / 2 b + 2 M i b $2 > 3$ „ $3 > 1$ y N , ^ :` b b K M M B V M

$$S_{231}(t; v) = x_1x_2 \quad x_1y_3 \quad x_2y_3 + y_2y_3$$

ÿ i ƒ , ^ L Ÿ y 0 ¥ ð 7 ^ = Q [T i i ƒ ž ^ Ð :` b b K M M B ¥ M É u Y b

---

[Rj] › œ w • n ¸ c b 2 + i B k Q X M j

| $v = 312$ 9 ^ d : ` b b K M M B M

$$S_{312}(t; i) = x_1 \quad t_3$$

» B „ Z 7 ð ¿ † [T Z 7

$$\begin{aligned} S_{231}(t; v)\ \ S_{312}(t; i) &= (x_1x_2 \quad x_1y_3 \quad x_2y_3 + y_2y_3)(x_1 \quad t_3) \\ &= x_1^2x_2 \quad x_1x_2t_3 \quad x_1^2y_3 + x_1y_3t_3 \\ &\quad x_1x_2y_3 + x_2y_3t_3 + x_1y_2y_3 \quad y_2y_3t_3 \end{aligned}$$

Ʊ œ Z 7 V

» = „ | Ä [• î t_j y_i ¥ ™ T
æ ¨ š © T $t_3 = (t_3 \quad y_3) + y_3$ | c t_3 ¥ [· s

$$\begin{aligned} x_1x_2t_3 &= \quad x_1x_2[(t_3 \quad y_3) + y_3] = \quad x_1x_2(t_3 \quad y_3) \quad x_1x_2y_3 \\ x_1y_3t_3 &= x_1y_3[(t_3 \quad y_3) + y_3] = x_1y_3(t_3 \quad y_3) + x_1y_3^2 \\ x_2y_3t_3 &= x_2y_3[(t_3 \quad y_3) + y_3] = x_2y_3(t_3 \quad y_3) + x_2y_3^2 \\ y_2y_3t_3 &= \quad y_2y_3[(t_3 \quad y_3) + y_3] = \quad y_2y_3(t_3 \quad y_3) \quad y_2y_3^2 \end{aligned}$$

» Ø „ † i] Ë [
| î µ c $(t_3 \quad y_3)$ ¥ [4

$$[\ x_1x_2 + x_1y_3 + x_2y_3 \quad y_2y_3]\ (t_3 \quad y_3)$$

» ¹ „ • î $c^w_{u;v}$ S_w ¥ ™ T
° a_3 † × a + ? m # 2 T V è R X d

w	$\ell(w)$	$S_w(t)$
123	0	1
213	1	x_1
132	1	$x_1 + x_2$
231	2	x_1x_2
312	2	x_1x_2
321	3	$x_1^2x_2$

| Z 7 ² T Ð $\{S_w g\}$ 1 V £ Ä [" " û ^ t_j y_i ¥ d µ œ " " " _ [T b
² , Ù – d : ` b b K M M † B ¥ Ø 7 ÷ ¯ Œ Y V] " ¥ ¥ X / F î µ " " ¯
– û ž Ÿ i i ƒ ž ^ ç Ø R X ¥ k , • ý b
6 B ñ ª ü Ÿ è 0 | $u = 321$ K É Â Ð $v = 123$ š © Â Ð b ® ¿

$$S_{321}(t; v) = (x_1 \quad y_1)(x_2 \quad y_2)(x_3 \quad y_3); \quad S_{123}(t) = 1$$

ð ü ^ $S_{321}(t; v)$ ' & Z 7 " " A – ^ 1 2 N b

k X 8 X E B ` % H H Q p K Ð $v = \ i = 0$ X

è R X R w , U R X R a F 2 ' µ Ø 0 ¥ ž Ÿ V X w , R X R ç Ø R X k $v = \ i = 0$ H ¥ +
è N H Z 7 " " | Ä ¹ ô Y d µ œ " b
V B î ™ T ž + y f ™
® ç Ø R X ¥ k Z 7 T

$$S_u(t; v)\ \ S_v(t; i) = \sum_w c^w_{u;v}(v; i)\ \ S_w(t; i)$$

7 v= i = 0 ¤

$$S_u(t;y)\ S_v(t;y) = \sum_w c^w_{u;v}(0;0)\ S_w(t;y)$$

Ï $c^w_{u;v}(0;0) \in N$ ^ ô Y d µ œ ” b

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î [w , R X@RN ¿ ý ð $S_u(t;y)\ S_v(t;y)$ a + ? m # 2 ¥ L Z 7 " " û ^ d µ œ ” b

› œ 9 Ø è 0 | $u = 231$ $v = 132$ b

» B „ p $S_{231}(t;y)$ „ $S_{132}(t;y)$

° V ¤

$$S_{231}(t;v) = x_1x_2 \quad x_1y_3 \quad x_2y_3 + y_2y_3; \quad S_{132}(t;i) = x_2 \quad t_2$$

7 v= i = 0 ¤

$$S_{231}(t;y) = x_1x_2; \quad S_{132}(t;y) = x_2$$

» = „ 9 Ø ð

$$x_1x_2\ \ x_2 = x_1x_2^2$$

» Ø „ • î S_w ¥ L Ÿ F †

° a_3 † × a + ? m # V 2 ` i

$$S_{123}(t) = 1; \quad S_{231}(t) = x_1x_2; \quad S_{321}(t) = x_1^2x_2$$

Z 7 " " ¹ 1 2 N ‹ S_{231} ¥ ñ ” £ $@_{w/v}(S_u) \in N[t]$ b

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| $u = 231$ $v = 132$ ñ d : ` b b K M M† B M

$$S_{231}(t;v) = x_1x_2 \quad x_1y_3 \quad x_2y_3 + y_2y_3$$

$$S_{132}(t;i) = x_2 \quad t_2$$

B î f ™ / ç Ø R X k

$$\begin{aligned} S_{231}(t;v)\ \ S_{132}(t;i) &= (x_1x_2 \quad x_1y_3 \quad x_2y_3 + y_2y_3)(x_2 \quad t_2) \\ &= x_1x_2^2 \quad x_1x_2t_2 \quad x_1y_3x_2 + x_1y_3t_2 \end{aligned}$$

7 v= i = 0 w , R X R

$$S_{231}(t;y) = x_1x_2; \quad S_{132}(t;y) = x_2$$

$$S_{231}(t;y)\ \ S_{132}(t;y) = x_1x_2\ \ x_2 = x_1x_2^2$$

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Ï $x^\delta = x_1^{n-1}x_2^{n-2} \cdots x_n^0$ b

'„Æ yM ª áÌ¤ž ×ñ'iiƒž^ *? Ti2`ùkîð THP¨¥ý

çI jXjUx a+?m#2[iT VX × a+?m#2[iT çI¹

U99V $$S_w(x;y) = \partial^{(x)}_{w^{-1}w_0} \prod_{i+j \leq n} (x_i + y_j):$$

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jXRX :`QKQp@qBMii2M:`QKQp@qBii2M BMXp VB^MÁ¯@™ 2 $H_2(V;Z)$ ^wLË +m`p2 +bH:`QbKQp@qBi,iM2M

UjXkV $$h\,I^V_{0;m;}\, i : H(V;Z)^{m} \to Z$$

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ƒt,M K Ÿ1þØÏ¥‰êÆ, ªŸ$øìÄiiƒ»^"ÐVþØÐ&| •!¥»BÜÅÃèb

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çØ jXRI U0]ØÌ (RR h?2Q`2KI30) V]ØÌ QH (Fl_n)] ¿

UjXjV $$\mathbb{Z}[x_1;\dots;x_n;q_1;\dots;q_{n-1}]/(e_1;\dots;e_n);$$

Ï $e_i(x|q)$ ^ 0 © ëf" ®[/› T3î

UjX9V $$/2\begin{pmatrix} x_1+t & q_1 & 0 & \cdots & 0 \\ -1 & x_2+t & q_2 & \cdots & 0 \\ 0 & -1 & x_3+t & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & x_n+t \end{pmatrix} = t^n + e_1 t^{n-1} + \cdots + e_n:$$

’ $q_1 = \;\; = q_{n-1} = 0$ H Ųj X ÿ|Ä ¹ Ü Å]Ø Ì i i ƒ 8 C 0 ñ ' Ð Ü Å ñ
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+ … c l 0 © ë f ” e_i ‹ ¿ @ ™ Ï + ç 0 @ ™ ¥ Ë b ñ Ì ¥ , Ä í •
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$q = (q_1; \dots; q_{n-1})$ ^ 0 M b
ä † [T i Q T T Q H v M ç Q ¹K B H

$$\tilde{S}_{w_0}(x;y) := \prod_{i=1}^{n-1} {}_i(y_{n-i} \mid x_1; \dots; x_i); \qquad (\text{Ųj X 8 V})$$

Ï

$$ {}_k(t \mid x_1; \dots; x_k) := \sum_{j=0}^{k} t^{k-j} e_j(x_1; \dots; x_k \mid q_1; \dots; q_{k-1}) \qquad (\text{Ųj X e V})$$

^ 1 ¿ $x_1; \dots; x_k$ ¥ 0 © ë f ” ¥ 3 î f ” b
¹ I ¹ ¡ ô ä † [T ô$ ’ $q = 0$ H $\tilde{S}_{w_0}(x;y) = \prod_{i+j \le n}(x_i + y_j)$ ƒ ž ^ ‹ K É
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ç l j X 9 0U × a + ? m # 2 [Ti (R R . 2 } M B i B X Q M Ä 9ñ) X Ð w 2 S_n ç l
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$$\tilde{S}_w(x;y) = \partial^{(y)}_{w w_0} \tilde{S}_{w_0}(x;y); \qquad (\text{Ųj X d V})$$

Ï " µ Ø 0 $\partial^{(y)}_{w w_0}$ T ¨ ¿ y M b

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| Ä ¹ Ü Å Ø , b
£ ! $q = 0$ 5 ${}_i(y_{n-i} \mid x_1; \dots; x_i) = \prod_{j=1}^{i}(x_j + y_{n-i+j})$ 7

$$\prod_{i=1}^{n-1}\prod_{j=1}^{i}(x_j + y_{n-i+j}) = \prod_{i+j \le n}(x_i + y_j) = S_{w_0}(x;y):$$

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è jXRS3U¥ 0 × a+?m#2{Ti VX S_3 Ï 0 × a+?m#2{Ti1

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回答. 1. 几何直观：维数公式 $\dim N^-(w) = \ell(w_0) - \ell(w)$

N^- 是单位下三角矩阵群，维数为 $\binom{n}{2}$（下三角矩阵的非对角元有 $\binom{n}{2}$ 个自由参数）。作为簇，N^- 同构于 $\mathbb{C}^{\binom{n}{2}}$，因为每个元素可以唯一写为根子群的乘积：

$$N^- = \prod_{\alpha\in\Phi^-} U_\alpha$$

其中 $U_\alpha \cong \mathbb{C}$ 是对应负根 α 的单参子群。

$N^-(w) = N^- \cap wN^- w^{-1}$ 的维数取决于哪些负根 α 满足 $w(\alpha) > 0$（即 w 把负根变成正根）。关键在于：

$$N^-(w) = \prod_{\substack{\alpha\in\Phi^- \\ w(\alpha)>0}} U_\alpha$$

负根是 $\Phi^- = \{-\epsilon_i + \epsilon_j : 1 \le j < i \le n\}$，共 $\binom{n}{2}$ 个。$w(\alpha) > 0$ 的负根 α 个数等于 $|J(w)| = \binom{n}{2} - \ell(w)$，其中 $J(w) = \{(i,j) : i < j, w(i) < w(j)\}$ 是 w 的非逆序集。

因此：

$$\dim N^-(w) = \binom{n}{2} - \ell(w) = \ell(w_0) - \ell(w)$$

几何图像：N^- 本身同构于最大的相反 Schubert 胞腔 $B^- w_0B/B \cong \mathbb{C}^{\binom{n}{2}}$。而 $N^-(w)$ 是 N^- 的一个”切片”，只保留那些 w 不翻转的负根方向。

2. 代数同构： $N^-(w) \cong B^- wB/B$

构造同构映射：

$$\phi : N^-(w) \to B^- wB/B, \quad x \mapsto xwB/B$$

验证是良好定义的：若 $x \in N^-(w)$，则 $x \in N^- \subseteq B^-$，所以 $xwB \in B^- wB$，陪集 xwB/B 落在 $B^- wB/B$ 内。

单射性：若 $xwB/B = ywB/B$，则 $w^{-1}x^{-1}yw \in B$。同时 $w^{-1}x^{-1}yw \in w^{-1}N^- w \cap N^- = N^-(w)$。但 $B \cap N^-(w) = \{e\}$，所以 $x = y$。

满射性：任取 $B^- wB/B$ 中元素。由于 $B^- = N^- T$，任意元素可写为 $n_1twB = n_1w(w^{-1}tw)B$。但 $w^{-1}tw \in T \subset B$，所以实际上 $n_1twB = n_1wB$，且 $n_1 \in N^-$，$n_1 \in N^-(w)$。

3. 同构映射 $x \mapsto xwB/B$ **的意义**

G/B 是齐性空间，$B^- wB/B$ 是过基点 wB/B 的胞腔。映射 $x \mapsto xwB/B$ 把 $N^-(w)$ 中的元素 x 看作”旗的位移”——把基旗 wB/B 推到旗 xwB/B。

$N^-(w) \cong \mathbb{C}^{\ell(w_0)-\ell(w)}$ 提供了胞腔 $B^- wB/B$ 的自然坐标系。每个 $x \in N^-(w)$ 指定了胞腔中一个唯一点。

从齐性空间角度看，wB/B 的稳定化子是 B，而 $N^-(w)$ 是那些”与 w 的相对位置在 N^- 中”的元素构成的子群。

4. 连通性

N^- 是幂幺群（unipotent group），指数映射 $\exp : \mathfrak{n}^- \to N^-$ 是双射（代数群情形的对数映射）。因此 N^- 微分同胚于其李代数 $\mathfrak{n}^- \cong \mathbb{C}^{\binom{n}{2}}$，是连通的。

$N^-(w)$ 是 N^- 的闭子群，因此作为代数群也是连通的。

同构 $N^-(w) \cong \mathbb{C}^{\ell(w_0)-\ell(w)}$ 说明 $N^-(w)$ 是仿射空间。仿射空间是连通的（甚至单连通）。

参考文献：Humphreys, [9, Section 28]。[oo]

30. 幂幺根基 U 的含义

30.1. 问题. 在 Borel 子群分解 $B = T \ltimes U$ 中，U 称为”幂幺根基”，这是什么意思？

[oo] 问：为什么 $N^-(w) \cong B^- wB/B$？见附录 0H§_nj ©k。

30.2. 回答. 幂幺（Unipotent）的定义:

一个代数群元素是幂幺的，当且仅当它所有特征值都为 1（对于复代数群，这意味着它是单参数群的乘积，每个单参数群都是形如 $\exp(t \cdot N)$ 的指数，其中 N 是幂零矩阵）。

几何直观:

在 $\mathrm{GL}_n(\mathbb{C})$ 中，幂幺矩阵的形式为上三角矩阵，对角线元素全为 1:

$$U = \left\{ \begin{pmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix} \right\};$$

这种矩阵称为幂幺，因为 $(u - I)^n = 0$（幂零）。

在一般约化群中:

设 G 为连通复约化群，$B \subset G$ 为 Borel 子群，$T \subset B$ 为极大环面。则

$$B = T \ltimes U, \quad B^- = T \ltimes U^-$$

其中 U 和 U^- 分别是包含在 B 和 B^- 中的极大幂幺子群。

为什么叫"根基"（Radical）?:

在群论中，radical（根基）是最大的可解理想。幂幺根基 U 是 B 的一个闭子群，它是 B 的"尾部"——移除了 torus T 的部分后，剩下的就是幂幺群。U 和 U^- 分别是对应于正根和负根的方向。

在旗流形中的作用:

在旗流形 G/B 中，Schubert 胞腔 $B \cdot wB/B$ 可以通过幂幺群 U^- 的作用参数化。具体而言，每个 $u \in U^-$ 给出一个群作用 $u \cdot (wB/B) = uwB/B$，这正是胞腔上的坐标。

与旗的变化的关系:

当你看到一个旗被"提升"时，实际上是 U^- 中的元素在作用。例如在 GL_n 中，U^- 是下三角幂幺矩阵，它们把标准旗变成一般旗。

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